Vusumuzi Nkosi

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SUMMARY

Data science postgraduate student (STADIO, NQF 8) and **ALX Data Science graduate** who frames every analysis around the business decision it supports. In my STADIO capstone I built an XGBoost model that flags new investing-platform accounts likely to stall before their first deposit; on unseen data, contacting the riskiest 20% of sign-ups reaches **1.79× more stallers than random outreach**. Nine years of mathematics teaching developed strong statistical thinking and the ability to explain results clearly to non-technical stakeholders. Seeking data scientist or data analyst roles in South Africa or remote.

TECHNICAL SKILLS

Programming: Python, SQL, R (basic)

ML & Modelling: scikit-learn, XGBoost, imbalanced-learn, logistic regression, gradient boosting, K-means clustering, feature engineering, threshold selection, time-based validation, cross-validation

Statistics & Experimentation: hypothesis testing, A/B testing, McNemar's test, confidence intervals, regression analysis, time-series forecasting (statsmodels), SciPy

Data & Visualisation: pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau, Excel

Tools & Platforms: Git, GitHub, Jupyter Notebook, Google Colab, AWS SageMaker, Prompt Engineering, Generative AI

PROJECTS

Account Activation Prediction — STADIO Capstone (CAP182) Python, XGBoost, scikit-learn

github.com/vusumuzinkosi/cap182-capstone

- Predicted, within 24 hours of registration, which new accounts on a retail investing platform would not make a first deposit within 30 days, using only data available at scoring time to prevent target leakage.
- Validated the approach on the public UCI Bank Marketing dataset (41,188 rows): XGBoost outperformed a logistic-regression baseline (PR-AUC 0.49 vs 0.46; McNemar's test $p < 0.001$).
- Applied it to the client's 8-table extract (1,841 accounts) with a time-based 60/20/20 split, tuned regularisation for the small cohort, and set the decision threshold on validation data.
- On the unseen test set (ROC-AUC 0.73), contacting the riskiest 20% of sign-ups gave a 1.79× lift over random outreach (bootstrap 95% CI 1.48–2.12) and captured 35% of all stallers, framing the cut-off as an onboarding-capacity decision; recorded an 11-minute video presentation of the findings.

Current projects — in progress, published in stages github.com/vusumuzinkosi

- **Churn Prediction: Revenue at Risk** — ranking customers by churn risk and estimating the revenue at stake (scikit-learn).
- **Fraud Detection for Mobile Money** — a daily review queue sized to investigator capacity, built for highly imbalanced data (imbalanced-learn).
- **Customer Segmentation** — RFM features and K-means clusters to design targeted campaigns (scikit-learn).
- **A/B Test: New Landing Page** — a rollout recommendation backed by hypothesis tests and confidence intervals (SciPy, statsmodels).
- **Eskom Demand Forecasting** — week-ahead forecasts of South Africa's electricity demand (statsmodels).

WORK EXPERIENCE

Senior Lecturer — Mathematics & IT Feb 2017 – Present

Amajuba TVET College — Newcastle, KwaZulu-Natal

- Deliver mathematics and IT instruction to 100+ students annually across multiple cohorts, designing curriculum and assessments aligned with national standards.
- Analyse student performance data to identify at-risk learners and adjust teaching strategies, improving pass rates through targeted, data-driven interventions.
- Developed standardised assessment rubrics adopted across the mathematics department for consistent grading.
- Mentored students in IT competitions at regional and national level, with multiple teams advancing to national rounds.

Special Educator — Mathematics Jan 2012 – Mar 2016

Mdumiseni High School — KwaZulu-Natal

- Delivered mathematics lessons for diverse learner groups, tracking individual progress and adapting materials to different learning needs.

EDUCATION

Postgraduate Diploma in Data Science (NQF 8) 2026 (in progress)

STADIO Higher Education — Remote

- 120-credit qualification (SAQA ID: 122721); modules: Introduction to Data Science, Working with Data, Statistical Modelling, Applied Machine Learning, Capstone Project.

Data Science Programme Mar 2025 – Apr 2026

ALX Africa — Remote

Postgraduate Diploma in TVET 2024 – 2025

University of the Western Cape

BSc Mathematics & Computer Science 2019 – 2024

University of South Africa

BEd Mathematics & Computer Science 2012 – 2017

University of South Africa

CERTIFICATIONS

- Data Science Certificate — ALX Africa (2025–2026)
- Data Analyst Certificate — ALX Africa (2025): SQL, Power BI, A/B Testing, Regression Analysis
- AI Engineering Bootcamp — Zero To Mastery Academy (2025)
- Prompt Engineering Bootcamp — Zero To Mastery Academy (2025)